

01 AI foundations



02 Generative AI



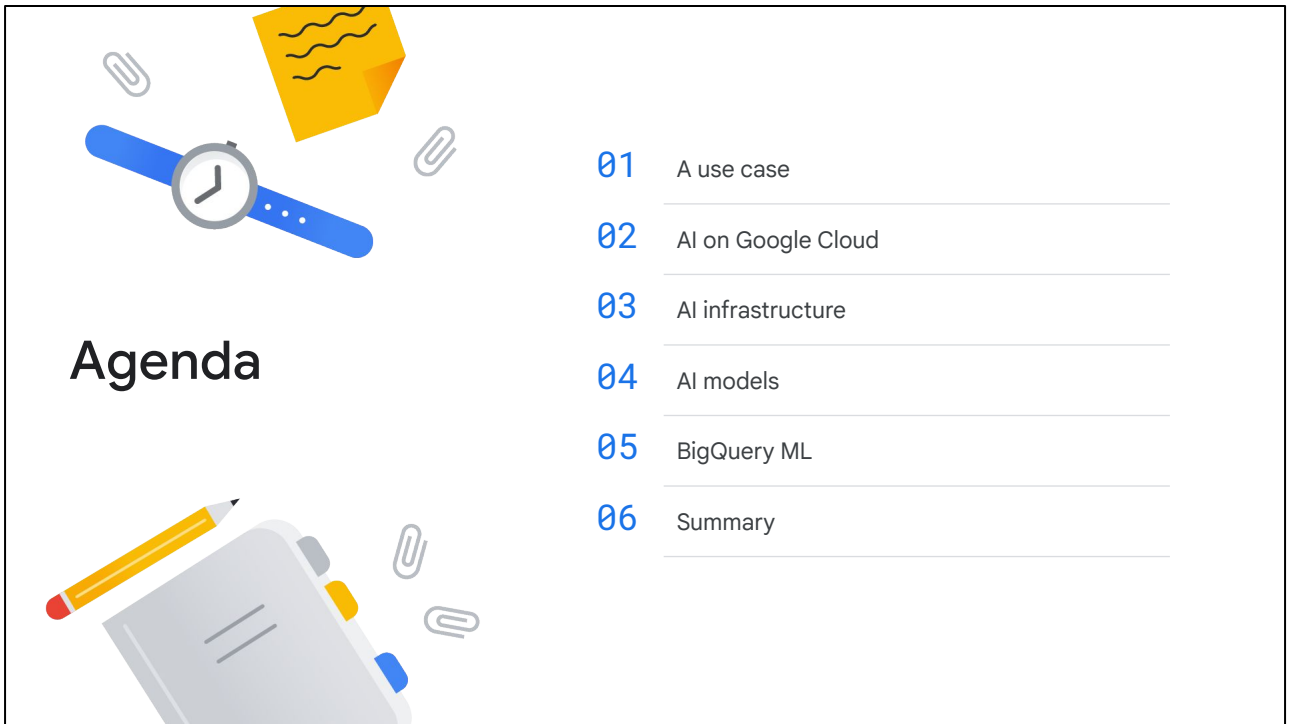
03 AI development options



04 AI development workflow



Welcome to the first module of this course, AI foundations.



01 A use case

02 AI on Google Cloud

03 AI infrastructure

04 AI models

05 BigQuery ML

06 Summary

You start your journey by exploring a real-world use case where AI addresses familiar daily challenges, revealing its profound capabilities and potential.

Next, understand how Google Cloud empowers your AI vision by solving both generative AI and traditional AI challenges with a comprehensive AI architecture, from infrastructure to development platforms, and then applications and solutions.

Then, a brief introduction to AI infrastructure will familiarize you with Google Cloud's compute, storage, data, and AI products.

Following this, delve into AI models, essential for understanding AI development.

Subsequently, get practical with a deep dive into BigQuery ML, connecting data and AI.

You conclude this module with a hands-on lab ,where you build your first ML model and leverage AI code assistance to explain, generate, and debug code.

AI is shifting from a tool for efficiency to a
powerhouse of innovation.

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Let's dive into a use case to see it in action.

Coffee on Wheels



Sells coffee on trucks in a similar way to a food stand.



Operates across cities like London.



Did you have your morning coffee today? If not, maybe you can use Coffee on Wheels, an international company that sells coffee on trucks in cities like London, New York, San Francisco, and Tokyo. They provide a compelling case study.

Challenges

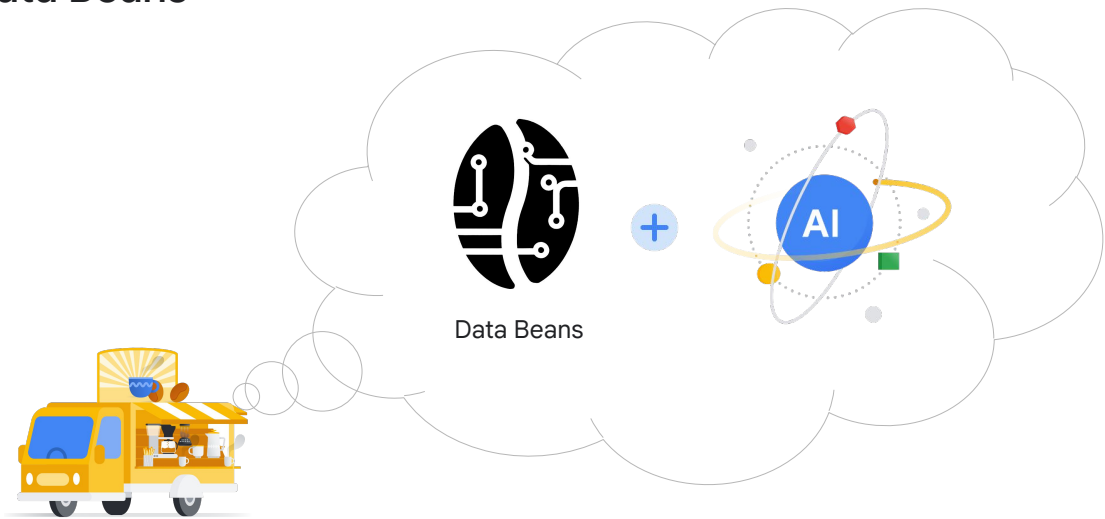


- ! Location selection and route optimization
- ! Sales forecast and real-time monitoring
- ! Marketing campaign automation

Coffee on Wheels is facing three main challenges:

- 1. Location selection and route optimization**
 - Predicting popular locations for truck placement and optimizing routes based on weather and traffic conditions.
- 2. Sales forecast and real-time monitoring**
 - Forecasting sales and monitoring performance in real-time.
- 3. Marketing campaign automation**
 - Automating marketing campaigns to increase efficiency and effectiveness.

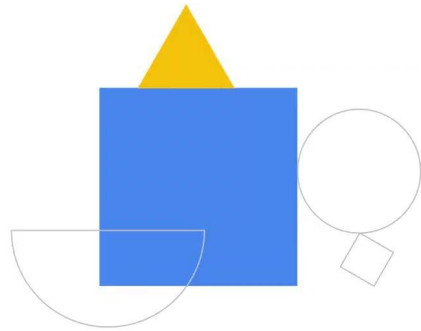
Data Beans



Recognizing the potential of AI, Coffee on Wheels sought assistance from Data Beans, a digital native company, to leverage data and AI technologies to resolve their business challenges.

Demo

Data Beans case study



Let's take a tour of the [demo](#).

Choose one of the four current locations, such as London. The dashboard displays overall statistics across cities, including revenue, operating margin, and the number of trucks. This information is generated by data tools like BigQuery and Looker, as well as AI tools and models like Gemini and Vertex AI.

On the right, you can view the final data for London with a summary. It shows how London's revenue compares to the average and provides insights into revenue per truck and customer loyalty.

In the top left corner, the dashboard displays the weather and generates route suggestions based on weather conditions. For example, if lower temperatures are forecasted, it might suggest a new itinerary that focuses on covered areas. You can click **show updated route** and **publish route** to implement these changes.

By clicking on a specific time on the timeline, you can see route suggestions based on city events. For example, if there's a football game happening, it might suggest rerouting trucks to avoid congestion.

Clicking on the truck sign provides a detailed dashboard with information such as

street view and revenue forecast.

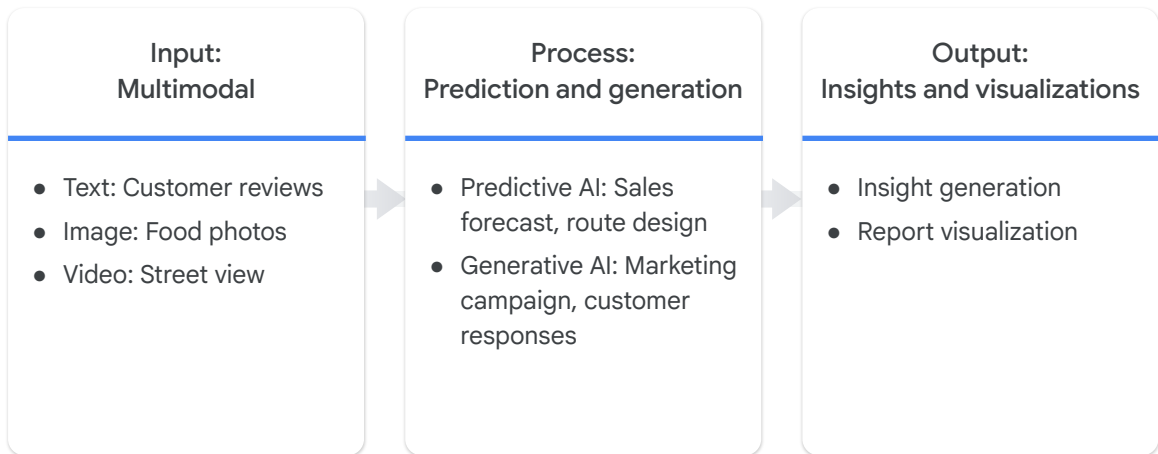
To monitor the performance of the business in real-time, you can access a dashboard by clicking **show menu**. If an item is underperforming, you can click the **generate** button to get suggestions for a new item.

Additionally, you have the option to generate marketing campaigns by selecting **yes** to save the suggestion. This feature enables you to automatically create campaigns that include both text and images. You can further streamline your marketing efforts by sending campaign emails to targeted customers with just a click of the **post** button.

Finally, you can generate an operational report and export the insights to any format, such as Google Slides.

To customize the application using code, click **how it's made**. This reveals the tools and technologies, including BigQuery, Gemini, and Vertex AI, that were used to create the app. You can also click **open in notebook** to access the sample code in the code development environment.

Unleash the power of AI

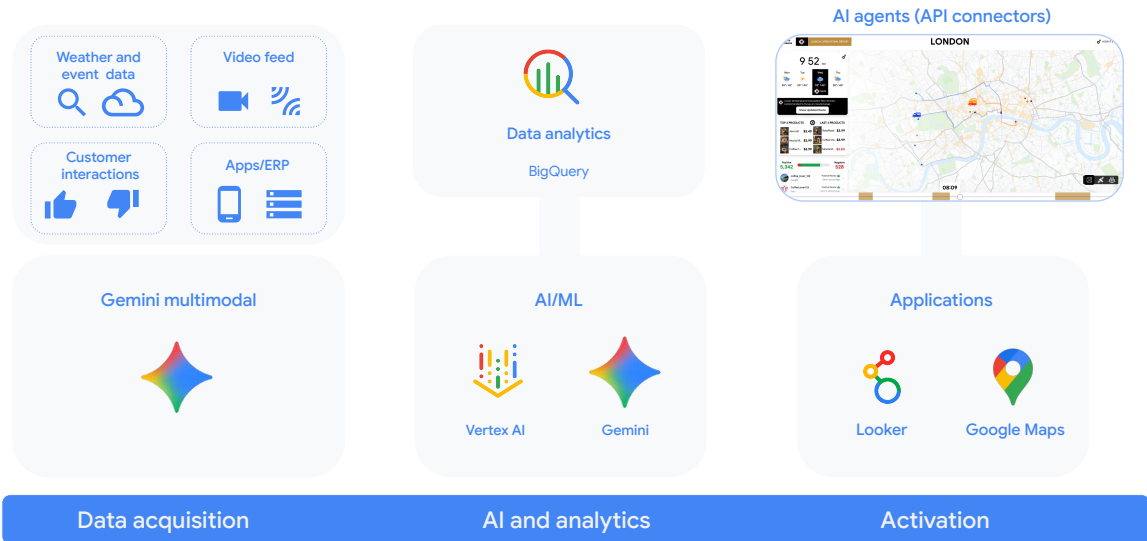


Isn't it remarkable?

The process is actually straightforward:

1. **Multimodal input:** This involves incorporating various forms of data, such as text (customer reviews), images (coffee and dessert pictures), and videos (real-time street view).
2. **Prediction and generation:** This is powered by predictive AI like sales forecasting, and generative AI like marketing campaign automation.
3. **Visual output:** The insights and reports are then presented visually, empowering businesses to make real-time data-driven decisions and optimize their operations.

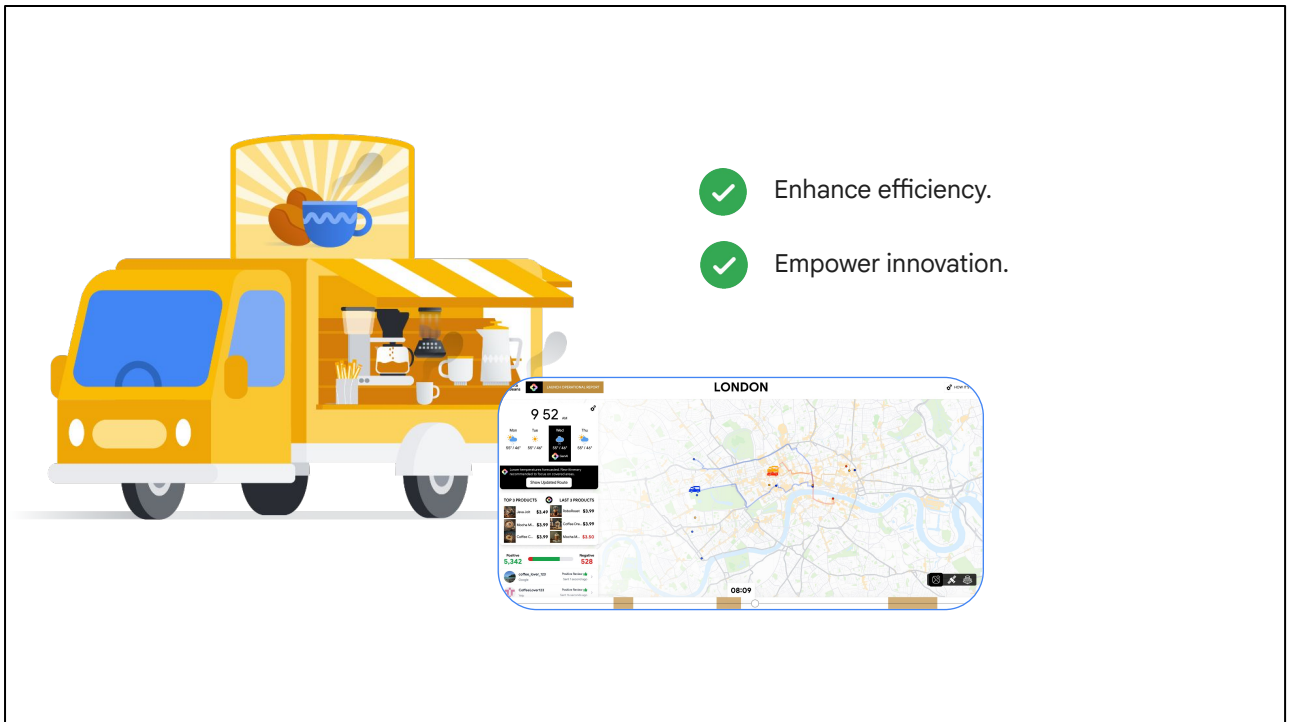
Data Beans' architecture



Behind the scenes, many Google products collaborate to make this application possible.

For example, Gemini multimodal enables data acquisition, BigQuery provides data analytics, Vertex AI handles ML development, and AI agents connect with various applications to enact decisions, such as interacting with Looker to visualize resulting insights.

You'll explore these tools in depth later in this course.



- ✓ Enhance efficiency.
- ✓ Empower innovation.

By leveraging the application, Coffee on Wheels **enhanced efficiency** by streamlining and automating business processes. This opened the door for **continuous innovation** in providing customers with immersive coffee experiences.

You can do the same for your business by using AI. Let's find out how Google can help you achieve your AI vision in the next lesson.



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Inspired by the Coffee on Wheels use case? Let's explore how Google can help you turn your own AI-powered ideas into reality.



History of leveraging AI to power products:

- Google Search
- Google Maps
- Workspace



Leader in AI and ML innovations:

- Gemini
- Vertex AI
- NotebookLM



Belief in responsible AI to foster:

- Bold innovation
- Responsible development and deployment
- Collaborative progress

So, why Google?

First, Google has a **long history** of leveraging AI to power its products, from Google Search to popular tools like Google Maps and Workspace. Google is eager to share its experience to empower you, whether an individual or an organization, in realizing your AI ambitions.

Second, Google **leads in AI and ML innovations**, particularly with recent generative AI breakthroughs. Technologies and products like Gemini, Vertex AI, and NotebookLM exemplify Google's commitment to delivering powerful AI services for your projects.

Third, Google believes in **responsible AI**. Google wants to collaborate with you to foster bold innovation, responsible development and deployment, and collaborative progress. Explore Google's Responsible AI principles in the reading to learn more.

And while this all sounds fantastic, you may be wondering **what is the current landscape of AI problems?**



Predictive AI

✓ Prediction of sales forecasting and route optimization

Generative AI

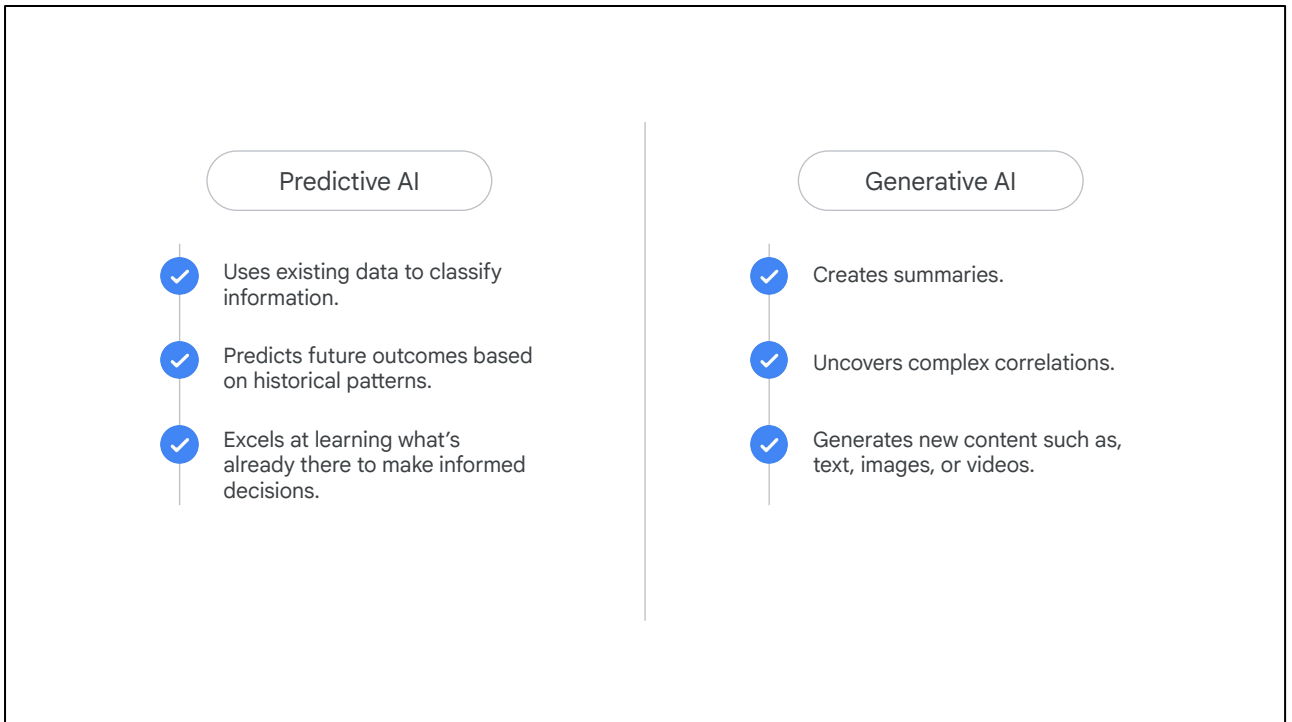
✓ Creation of customer responses and automation of marketing campaigns

To simplify, let's categorize the problems.

Recall the "Coffee on Wheels" use case presented earlier in this module. What problems could AI solve for them?

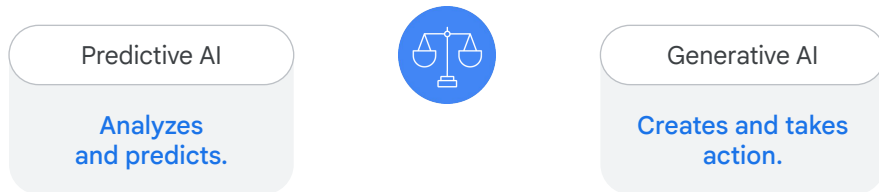
- One type of problem is **prediction**, exemplified by sales forecasting and route optimization through traffic prediction. This is known as predictive AI.
- The other type is **creation**, such as generating customer responses and automating marketing campaigns. This is called generative AI.

Let's delve deeper into predictive and generative AI, comparing them and determining their optimal applications.



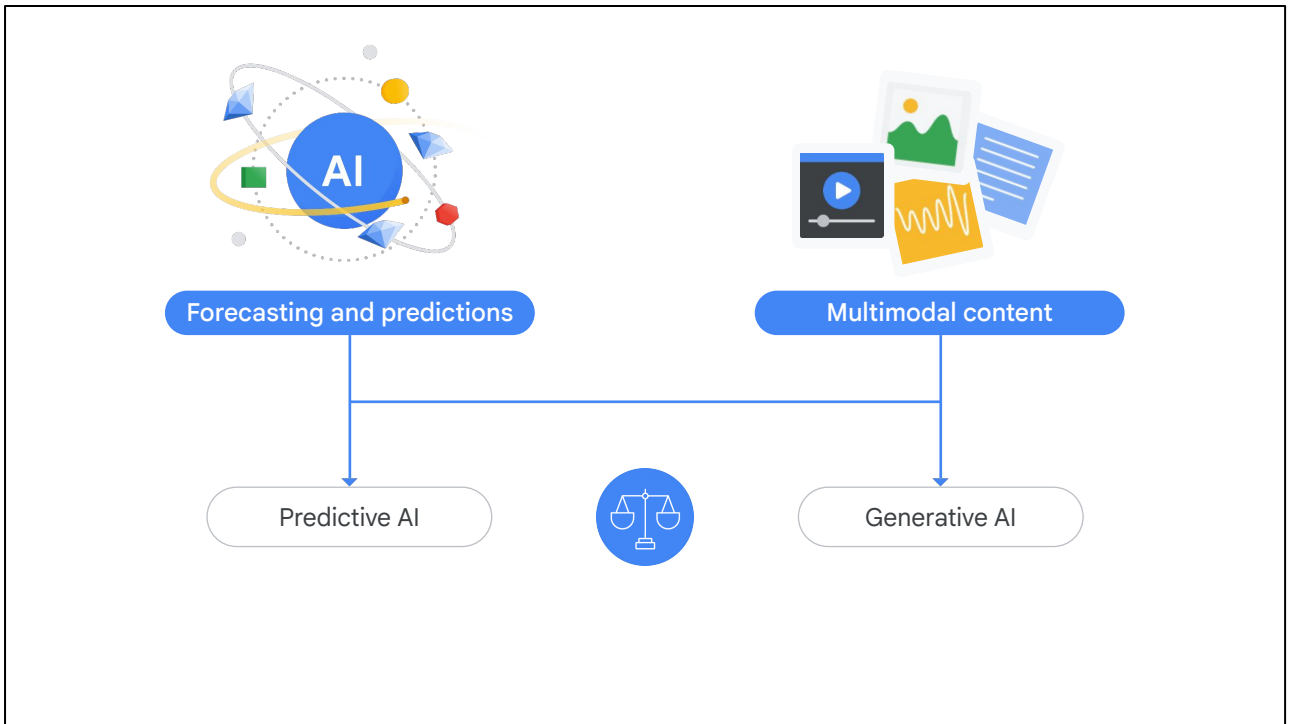
So what is predictive AI and generative AI?

- **Predictive AI** (also known as transitional or discriminative AI) uses existing data to classify information or predict future outcomes based on historical patterns. It excels at learning from what's already there to make informed decisions, much like using tried-and-true methods to sort and forecast.
- **Generative AI**, on the other hand, expands these capabilities to create summaries, uncover complex correlations, or generate new content. This includes text, images, or videos that mirror the style and patterns within the training data. It doesn't just analyze; it creates!



To put it simply:

- Predictive AI: analyzes and predicts.
- Generative AI: creates new content and takes action.

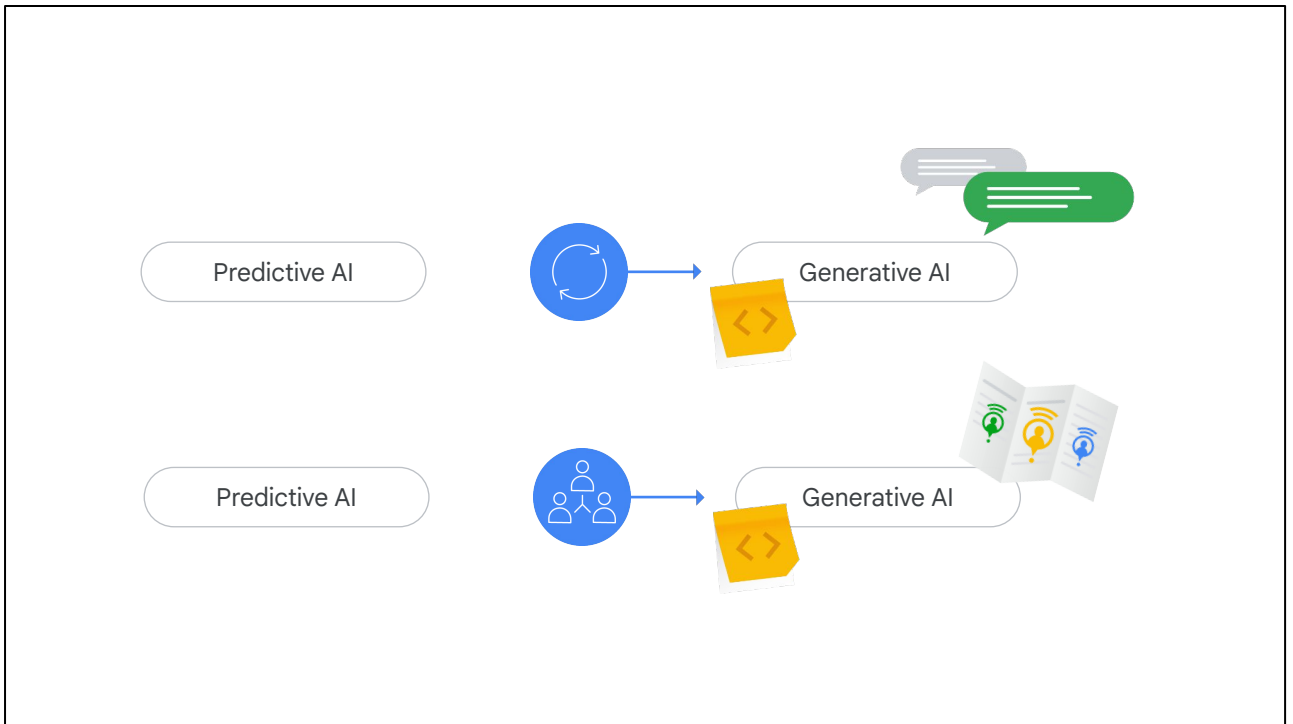


Now when you should use them is based on your use case.

Here's a simplified decision tree:

- For forecasting and predictions, a predictive AI model might be your go-to.
- And for multimodal content generation (text, image, video) and automation, generative AI is ideal.

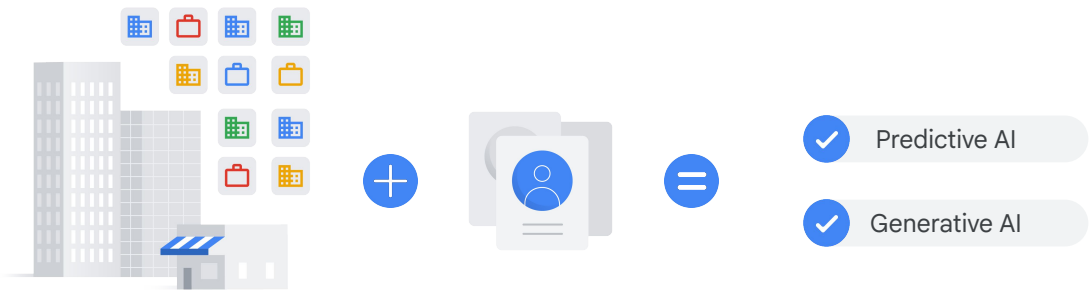
However, there's no clear line between these two, and sometimes the best approach is to use both!



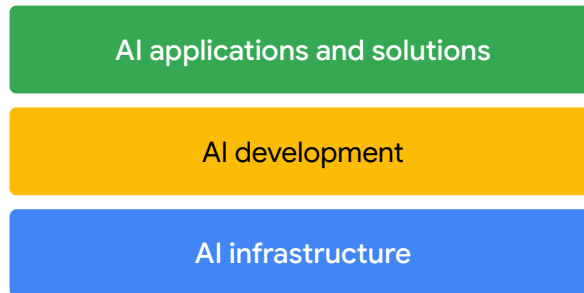
You can use the output from a **Predictive AI** model as part of the prompt for a Generative AI model.

For example,

- Use **Predictive AI** to forecast customer churn, then use **Generative AI** to power a chatbot that helps your sales team explore these predictions.
- Or Use **Predictive AI** to identify customer segments, then use **Generative AI** to create personalized marketing content for each segment.



By prioritizing business outcomes and user needs, you can maximize the benefits of both types of AI.



You can think of the Google Cloud infrastructure in terms of three layers:

- It all starts with a robust AI infrastructure, featuring advanced compute, network, and storage technologies. This foundational layer provides the solid ground for building your AI realm.
- Next, we move up to the development layer, where the real magic happens. Google's Vertex AI, an end-to-end AI development platform, guides you from design to deployment. Powering Vertex AI are Google's foundational models like Gemini and streamlined deployment pipelines. Vertex AI also seamlessly integrates with data products like BigQuery, ensuring a smooth journey from data to AI. This AI development layer is truly the ultimate playground for developers, engineers, and data scientists.
- And what if you're not a technical professional? At the top layer of applications and solutions, Google provides out-of-the-box options for business users and analysts to rapidly prototype their ideas.



- Gen AI foundational models
- Gen AI tools for development and application levels

Generative AI
Module 2

Predictive AI
Module 3
Module 4

- Options to train ML models
- End-to-end ML model building

AI infrastructure
Module 1

Excited? You'll learn about AI infrastructure soon!



Agenda



01 A use case

02 AI on Google Cloud

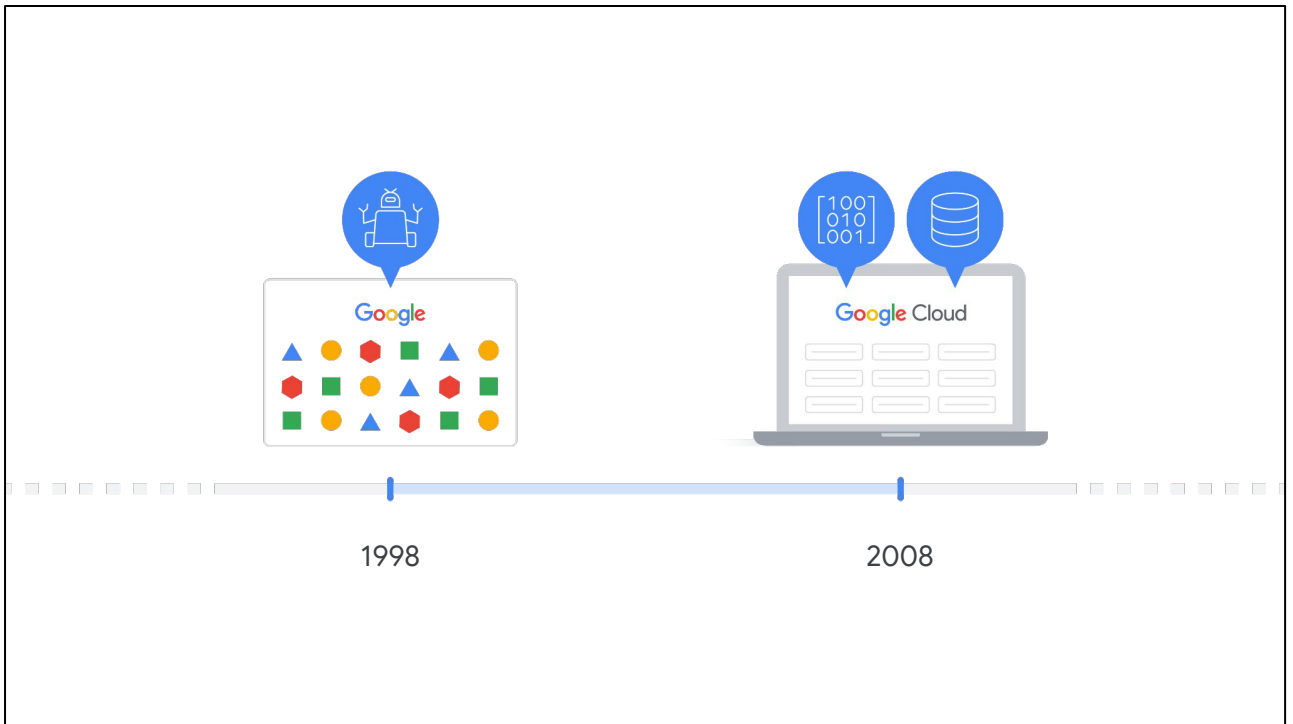
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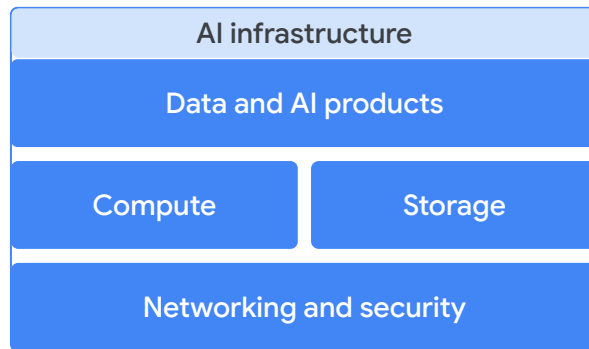
06 Summary

Building on the previous lesson, where you explored Google's three-layered AI architecture, including AI infrastructure, AI development, and AI applications and solutions, you'll now delve into the foundational layer: AI infrastructure.



Since its inception in 1998, Google has been dedicated to data and AI. A decade later, in 2008, Google Cloud was introduced to offer secure and flexible cloud computing and storage solutions.

Google Cloud



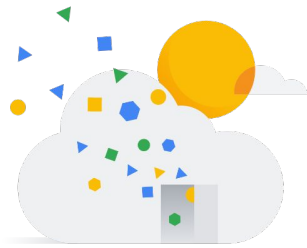
You can think of the Google Cloud infrastructure in terms of three layers:

- At the base layer is *networking and security*, which lays the foundation to support all of Google's infrastructure and applications.
- On the next layer sit *compute* and *storage*. Google Cloud separates, or decouples, as it's technically called, compute and storage so they can scale independently based on need.
- The top layer includes *data and AI products*, which enable you to perform tasks to ingest, store, process, and deliver business insights, data pipelines, and ML models.

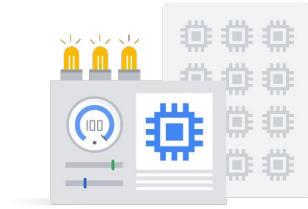
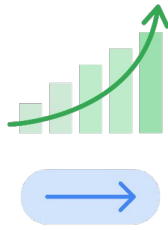
Thanks to Google Cloud technology, these tasks can be accomplished without needing to manage and scale the underlying infrastructure.

However, understanding some essentials about Google Cloud compute and storage can help you grasp the higher-level data and AI products.

Let's begin with *compute*.

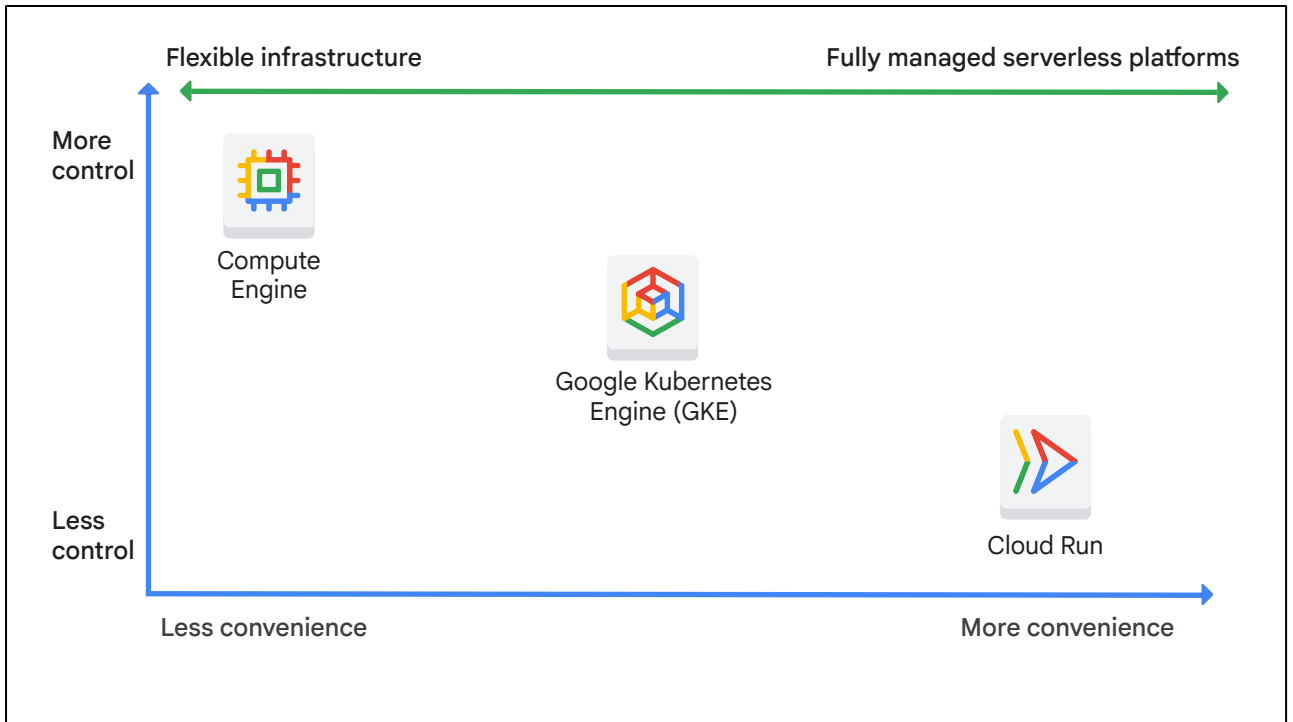


Data needs



Compute power

Organizations with growing data needs often require lots of compute power to run data and AI jobs. And as organizations design for the future, the need for compute power only grows.



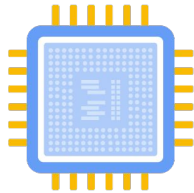
Google offers a range of computing services, from **flexible infrastructure** to **fully managed serverless platforms**, balancing **control** and **convenience**.

For example,

- **Compute Engine:** High control, like managing a physical server.
- **Google Kubernetes Engine (GKE):** Control over containerized apps with orchestration benefits.
- **Cloud Run:** Serverless convenience, Google manages infrastructure.

You might be familiar with the container platform GKE and serverless options like Cloud Run.

For more details, check out Google documentation in the reading list.



CPU

Central processing unit

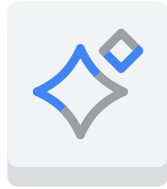
GPU

Graphics processing unit

Where does the processing power come from? It's from the hardware: computer chips.

However, traditional computer chips, like central processing units, or CPUs, and even the more recent graphics processing units, or GPUs, may no longer scale to adequately reach the rapid demand for AI.

2016

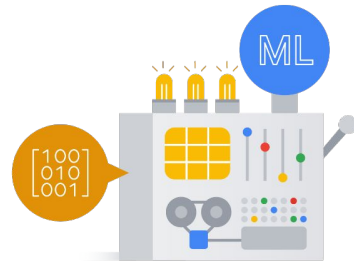
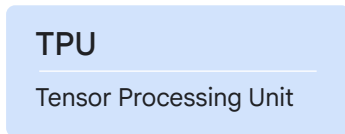


Tensor Processing Unit (TPU)

Google's customized application-specific chips to [accelerate AI workloads.](#)

in 2016, Google introduced the *Tensor Processing Unit*, or TPU. TPUs are Google's customized application-specific chips to accelerate AI workloads.

Domain-specific hardware



Allows for higher efficiency.

TPUs act as domain-specific hardware, as opposed to *general-purpose* hardware like CPUs and GPUs.

This allows for higher efficiency by tailoring the architecture to meet the computation needs in a domain, such as the matrix multiplication in machine learning.

TPUs are **faster** and more **energy-efficient** for AI/ML, available for Google Cloud customers.

Cloud TPUs, faster and more energy-efficient than GPUs and CPUs for AI/ML, are integrated across Google products, offering state-of-the-art supercomputing technology to Google Cloud customers.



Decoupled
Cloud computing



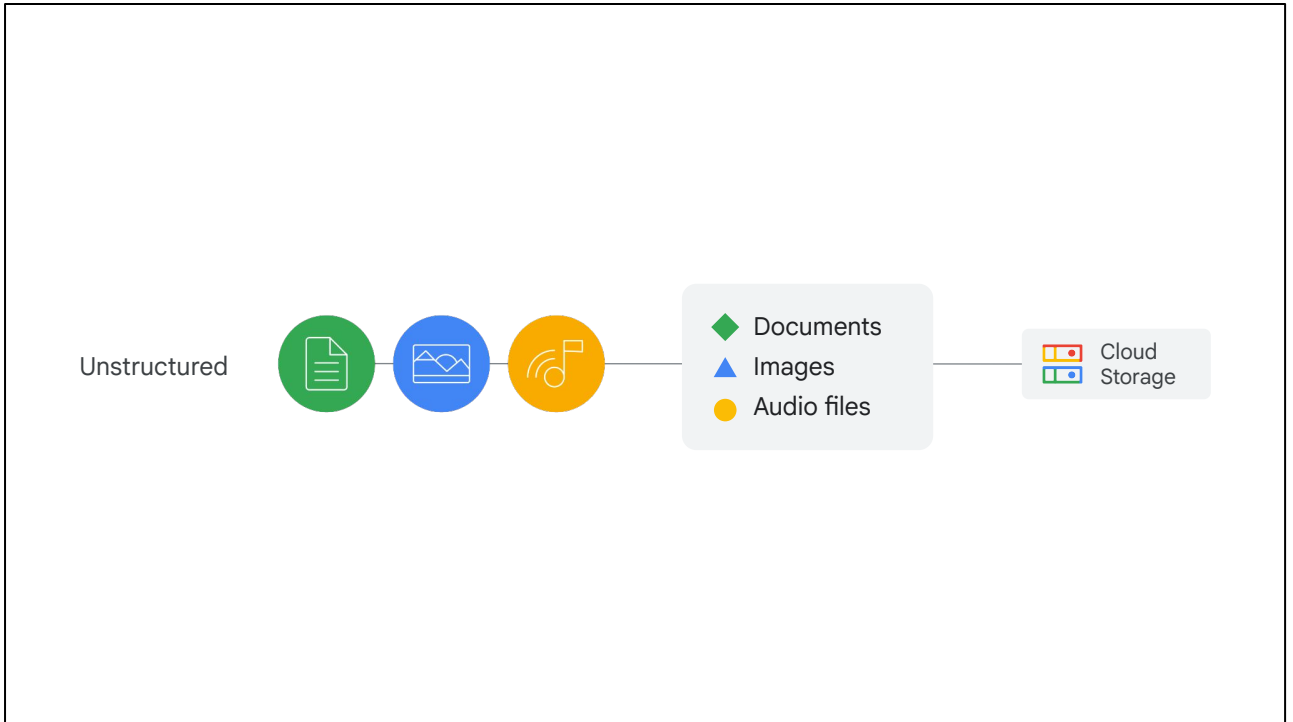
Coupled
Desktop computing

Let's now examine storage.

For proper scaling capabilities, compute and storage are decoupled.

That is one major difference between cloud and desktop computing.

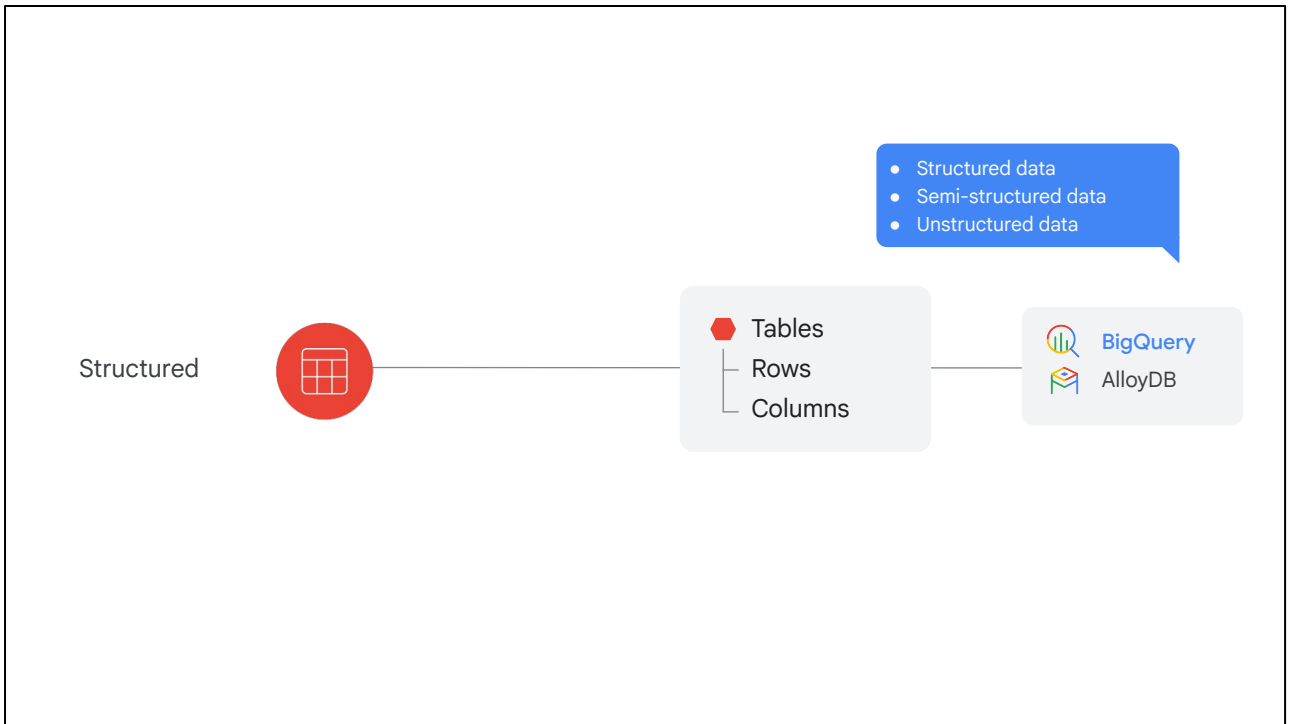
With cloud computing, compute and storage can scale separately.



Most applications need a database and storage solution of some kind.

Your best option depends on your data type and business needs.

For unstructured data like documents, images, and audio files, Cloud Storage is your ideal choice.



Alternatively, If your data is structured, organized in tables, rows, and columns, you have options like **BigQuery**, **AlloyDB for PostgreSQL**, and others.

Note that BigQuery, Google's flagship data warehouse, is particularly versatile. It's built for structured data and also highly optimized for semi-structured data like JSON.

It can even query unstructured data, such as log files or images stored in Cloud Storage, by creating an **external table** that provides a structured reference to that data.

Google Cloud

Data and AI products

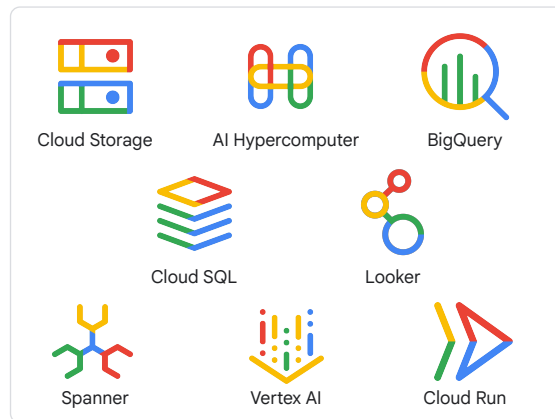
Compute

Storage

Networking and security

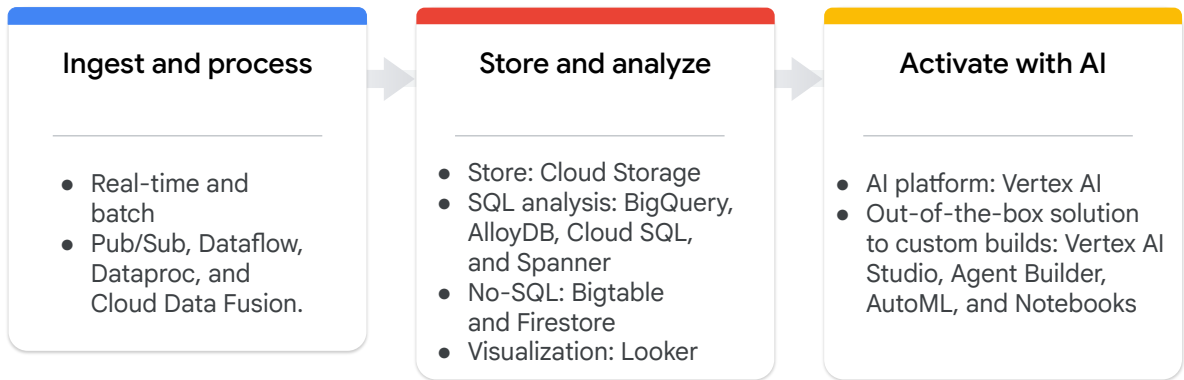
This leads to the top layer of the Google Cloud infrastructure, data and AI products.

Google Cloud —



As you explored earlier, Google Cloud offers a comprehensive suite of data and AI tools. How do you piece them together?

Data-to-AI workflow



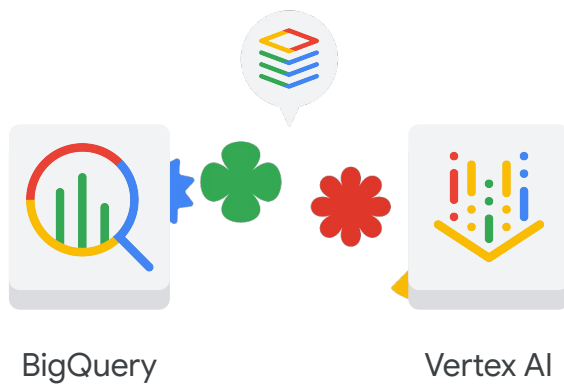
To build a data and AI project, you orchestrate these products through a data-to-AI workflow:

- Ingest and process
- Store and analyze
- and Activate with AI.

First, ingest and process data from diverse sources, both real-time and batch, using tools like Pub/Sub, Dataflow, Dataproc, and Cloud Data Fusion.

Next, store your data in solutions like Cloud Storage. Then, analyze it with various tools: BigQuery, AlloyDB, Cloud SQL, and Spanner for SQL databases; Bigtable and Firestore for NoSQL databases; and Looker for visualization.

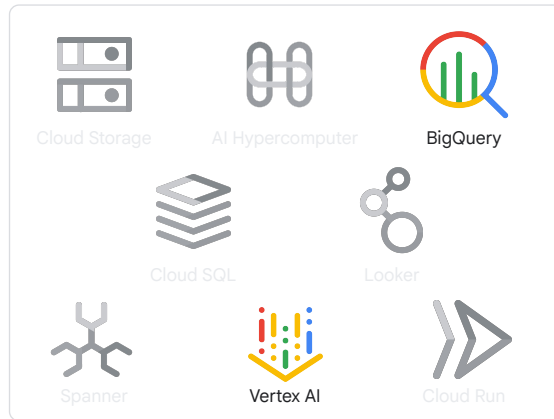
Finally, activate your insights with AI. Train predictive models for forecasting or leverage gen AI for content creation and action. Vertex AI is the central AI development platform, offering products like Vertex AI Studio, Agent Builder, AutoML, and Notebooks for AI projects ranging from out-of-the-box solutions to custom builds.



These tools are seamlessly integrated on Google Cloud, enabling data scientists and AI developers to efficiently transition from data to AI.

For example, BigQuery offers embedded SQL comments to train an ML model, a feature you'll explore later. Additionally, within a Vertex AI notebook, you can easily pull data directly from BigQuery using SQL for advanced model training.

Google Cloud —



Don't let the variety of options overwhelm you. You'll focus on BigQuery, the primary data warehouse, and Vertex AI, the AI development platform, later in this course. But, before that, let's get you ready with another fundamental topic, AI models, in the next lesson.



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01 A use case

02 AI on Google Cloud

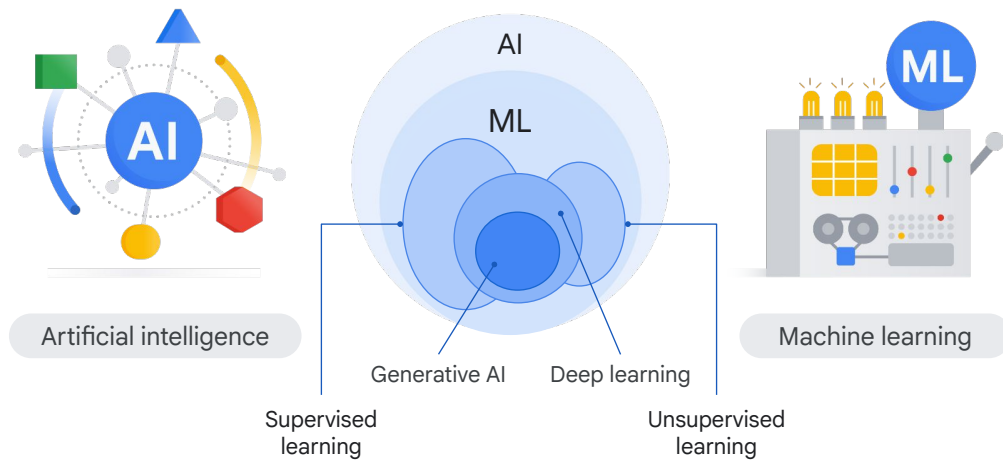
03 AI infrastructure

04 AI models

05 BigQuery ML

06 Summary

Before you dive into more practical topics and build your own machine learning model, let's prepare you with foundational knowledge and explore the AI model categories.



First, let's pause to clarify two terms: artificial intelligence and machine learning. You may note that people often use the terms interchangeably, but they do have some differences.

Artificial intelligence, or AI, is an umbrella term that includes anything related to computers mimicking human intelligence. Some examples of AI applications include robots and self-driving cars.

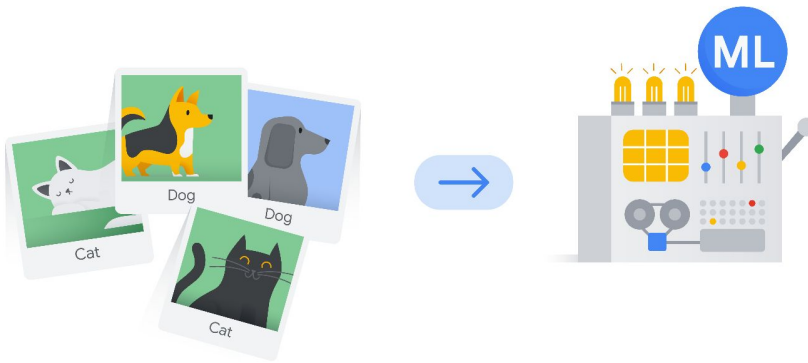
Machine learning is a subset of artificial intelligence (AI) that allows computers to learn without being explicitly programmed. This is in contrast to traditional programming, where the computer is told explicitly what to do. Machine learning mainly includes supervised and unsupervised learning.

You might also hear the terms deep learning, or deep neural networks. This is a subset of machine learning that adds layers in between input data and output results to make a machine learn at much depth. You'll learn more about neural networks and deep learning later in the course.

Finally, **generative AI** (gen AI) creates content and performs tasks based on requests. Gen AI uses foundation models, like large language models (a type of deep learning model), to predict, interpret, and interact with language.

You'll delve deeper into gen AI models in the next module.

Supervised learning

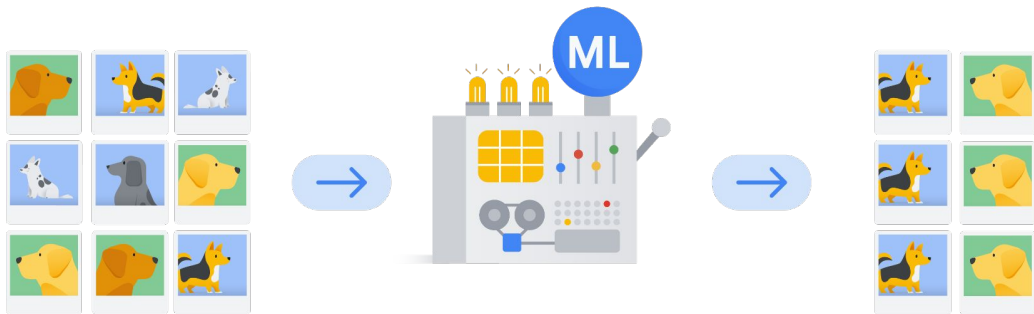


So, what's the difference between supervised and unsupervised learning?

Imagine two types of problems:

In problem one, you are asked to classify dogs and cats from a very large set of pictures. You already know the difference between dogs and cats. So you *label* each picture and pass the *labeled* pictures to a machine. By learning from the data, in this case, pictures with the answers or labels, *supervised learning is being enacted*, allowing the machine to tell if a new picture represents a dog or cat in the future.

Unsupervised learning



In problem two, you are asked to classify breeds of dogs. Unfortunately, this time you don't know many of them and are not able to label the pictures. So you send these *unlabeled* pictures to a machine. In this case, the machine learns from the data without the answers and finds underlying patterns to group the animals. This is an example of *unsupervised learning*.

Supervised learning



Labeled data, task-driven,
identifies a goal.

Unsupervised learning

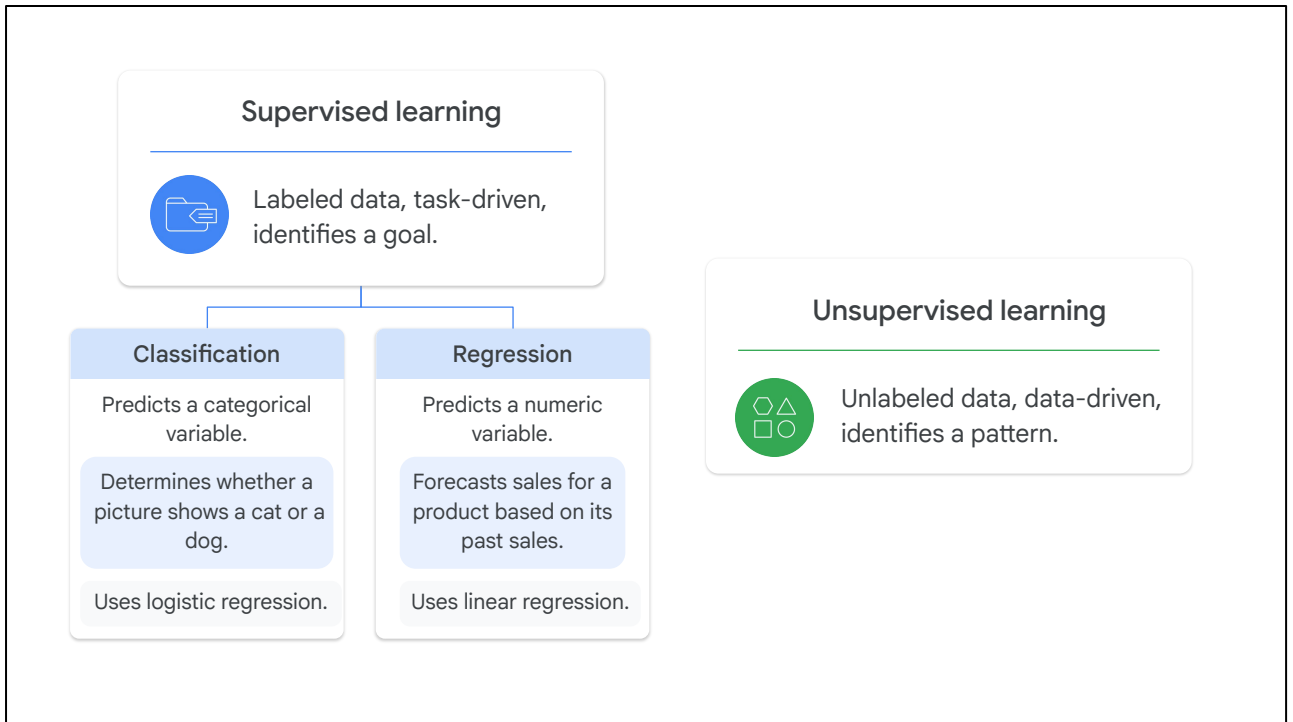


Unlabeled data, data-driven,
identifies a pattern.

Put simply, *supervised learning* deals with labeled data, is task-driven, and identifies a goal.

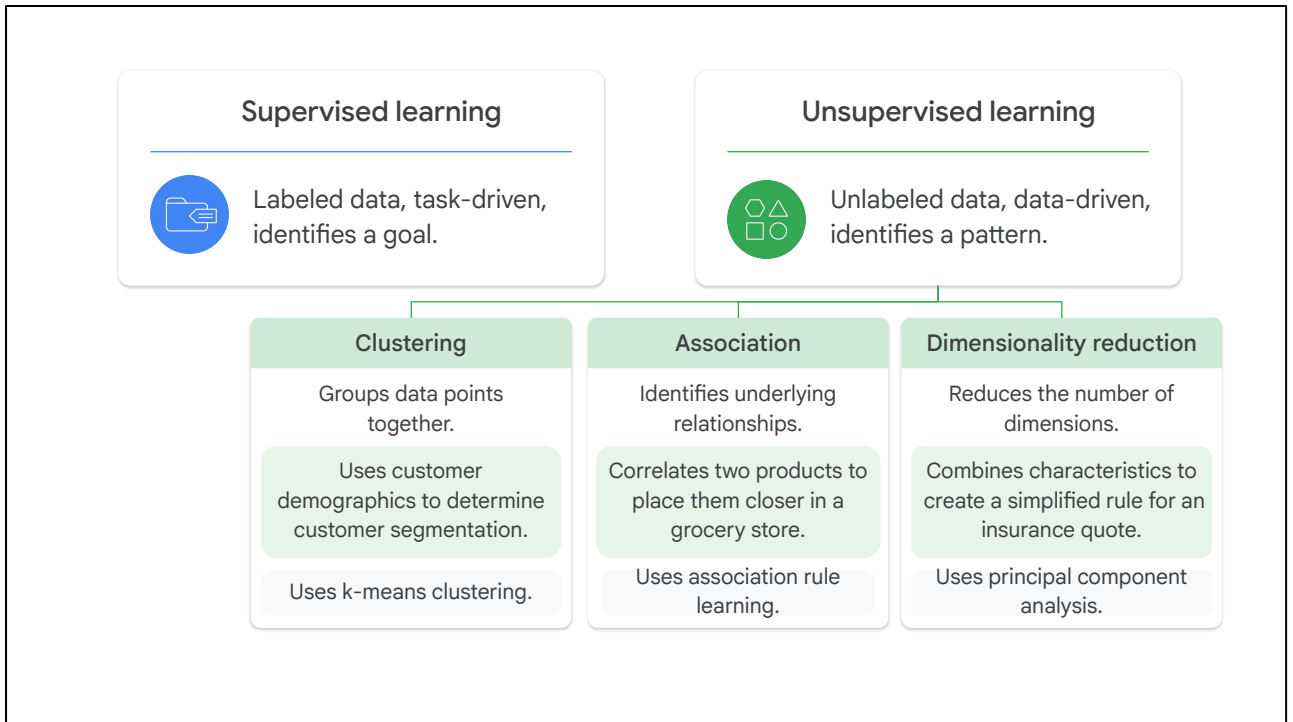
Unsupervised learning, however, deals with unlabeled data, is data-driven, and identifies a pattern.

An easy way to distinguish between the two is that supervised learning provides each data point with a label, or an answer, while unsupervised learning does not.



There are two major types of supervised learning:

- The first is **classification**, which predicts a categorical variable, such as determining whether a picture shows a cat or a dog. In ML, you use models like a logistic regression model to solve classification problems.
- The second type of supervised learning is **regression**, which predicts a numeric variable, like forecasting sales for a product based on its past sales. You use ML models like a linear regression model to solve regression problems.



There are three major types of *unsupervised* learning:

- The first is **clustering**, which groups together data points with similar characteristics and assigns them to "clusters," like using customer demographics to determine customer segmentation. You use ML models like k-means clustering to solve clustering problems.
- The second type is **association**, which identifies underlying relationships, like a correlation between two products to place them closer together in a grocery store for a promotion. You use association rule learning techniques and algorithms like Apriori to solve association problems.
- And the third type of unsupervised learning is **dimensionality reduction**, which reduces the number of dimensions, or features, in a dataset to improve the efficiency of a model. For example, combining customer characteristics like age, driving violation history, or car type, to create a simplified rule for calculating an insurance quote. You use ML techniques like principal component analysis to solve these problems.

Example 1

Predict customer spending based on purchase history.

Date	Customer ID	Age	Gender	Address	Amount
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x

All right, time to test your learning.

You are asked to predict customer spending based on purchase history.

- Is this supervised or unsupervised learning?
- Is this a classification or a regression problem?
- Which ML model should you use? A logistic regression or a linear regression?

Example 1

Predict customer spending based on purchase history.

Date	Customer ID	Age	Gender	Address	Amount
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x



Supervised learning



Regression problem



Linear regression

Did you get the answers right?

- Yes, that's *supervised* learning because you have the labeled data, the amount the customers have spent, and you want to predict their future purchases.
- Yes, it's a *regression* problem because it predicts a continuous number: future spending.
- Yes, a *linear* regression. A logistic regression model is for classification problems, while a linear regression model is for regression problems.

Example 2

Identify customer segmentation.

Date	Customer ID	Age	Gender	Address	Amount
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x

Let's look at another scenario.

Imagine you are using the same dataset. However, this time you are asked to identify customer segmentation. You don't want to base your judgement on stereotypes such as age or gender, so you use a computer for help.

- Is this supervised or unsupervised learning?
- Is it a clustering, association, or dimensionality deduction problem?
- Which ML model should you use: logistic regression, linear regression, or k-means clustering analysis?

Example 2

Identify customer segmentation.

Date	Customer ID	Age	Gender	Address	Amount
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x
x	x	x	x	x	x



Unsupervised learning



Clustering problem



Clustering analysis

Did you get the answers right?

- Yes, it's *unsupervised* learning because you don't have each customer labeled as belonging to a certain segment. Instead, you want the computer to discover the underlying pattern.
- Yes, identifying customer segmentation is a *clustering* problem.
- Right, it's a *clustering analysis* scenario. You will find these models within BigQuery ML, AutoML, and custom training later on in this course.



Agenda



01 A use case

02 AI on Google Cloud

03 AI infrastructure

04 AI models

05 BigQuery ML

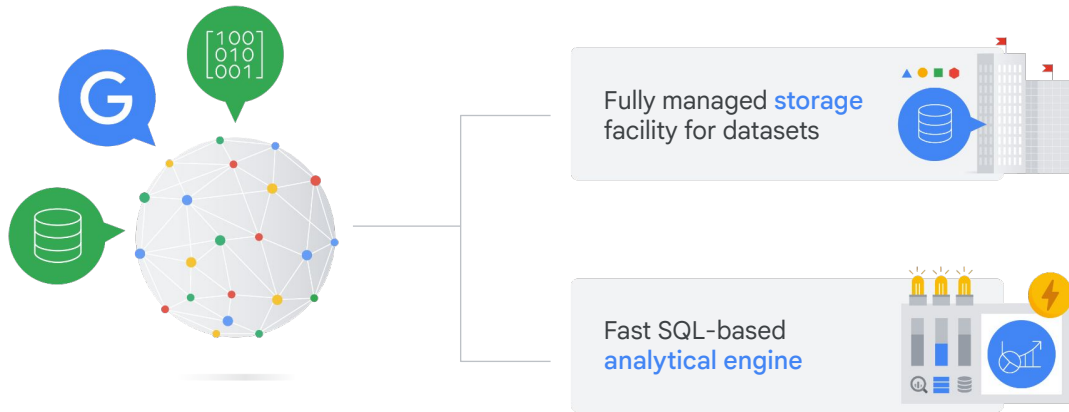
06 Summary

With the different types of ML models in your mind, let's apply concept to practice. In this lesson, you explore BigQuery ML and walk through the steps to build an ML model with SQL commands.



You learned about BigQuery, the primary data analytics tool on Google Cloud, from the previous lesson.

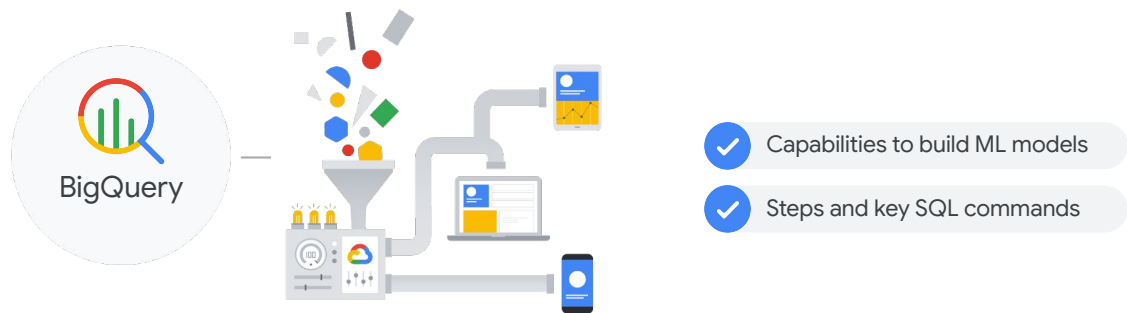
BigQuery = Storage + analytics



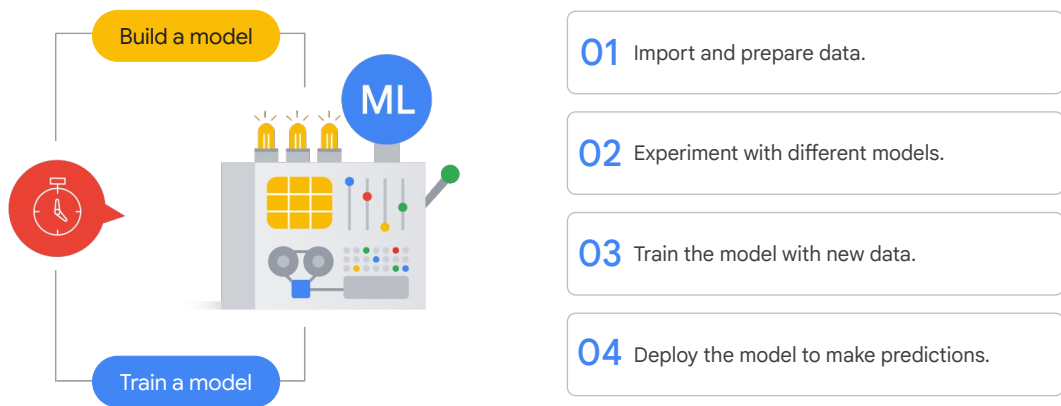
You learned about BigQuery, the primary data analytics tool on Google Cloud, from the previous lesson.

BigQuery provides two services in one. It's a fully managed *storage* facility to load and store datasets, and a fast SQL-based *analytical engine*.

The two services are connected by Google's high-speed internal network. It's this super-fast network that allows BigQuery to scale both storage and compute independently, based on demand.



Although BigQuery started out solely as a data warehouse, over time, it has evolved to provide features that support the data-to-AI lifecycle, meaning you can perform both data analytics and build pre-defined ML models within BigQuery. In this lesson, you explore BigQuery's capabilities to build ML models and walk through the steps and key SQL commands to do so.



If you've worked with ML models before, you know that building and training them can be very time-intensive.

- You must first import and prepare the data.
- Then, experiment with different ML models and tune the parameters.
- To improve model performance, you also need to go back and forth to train the model with new data and features.
- And finally you need to deploy the model to make predictions.

This is an iterative process that requires a lot of time and resources.



Now, with **BigQuery ML**, you can **manage**
the **tabular data** and **execute ML models**
in one place with just a few steps.

Now, with BigQuery ML, you can manage tabular data and execute ML models in one place with just a few steps. BigQuery ML tunes the parameters for you and helps you manage the ML workflow.

Key phases

01 Extract, transform, and load data into BigQuery.

02 Select and preprocess features.

03 Create the model inside BigQuery.

04 Evaluate the performance of the trained model.

05 Use the model to make predictions.

If you're already using other Google products, look out for easy connectors to get that data into BigQuery before you build your own pipeline.

You can enrich your existing data warehouse with other data sources by using SQL joins.

Let's walk through the phases of a machine learning project and the key SQL commands.

In *phase 1*, you *extract, transform, and load data into BigQuery* if it isn't there already.

If you're already using other Google products, like YouTube for example, look out for easy connectors to get that data into BigQuery before you build your own pipeline.

You can enrich your existing data warehouse with other data sources by using SQL joins.

Key phases

01 Extract, transform, and load data into BigQuery.

02 Select and preprocess features.

Use SQL to create the training dataset for the model to learn from.

03 Create the model inside BigQuery.

BigQuery ML does some of the preprocessing for you, like one-hot encoding of your categorical variables.

04 Evaluate the performance of the trained model.

05 Use the model to make predictions.

In *phase 2*, you *select and preprocess features*. You can use SQL to create the training dataset for the model to learn from.

BigQuery ML does some of the preprocessing for you, like one-hot encoding of your categorical variables. One-hot encoding converts your categorical data into numeric data that is required by a training model.

Key phases

01 Extract, transform, and load data into BigQuery.

02 Select and preprocess features.

03 Create the model inside BigQuery.

04 Evaluate the performance of the trained model.

05 Use the model to make predictions.

Use the `CREATE MODEL` command.

```
#standardSQL
CREATE MODEL
ecommerce.classification

OPTIONS
(
  model_type='logistic_reg',
  input_label_cols =
    'will_buy_later'
) AS

# SQL query with training data
```

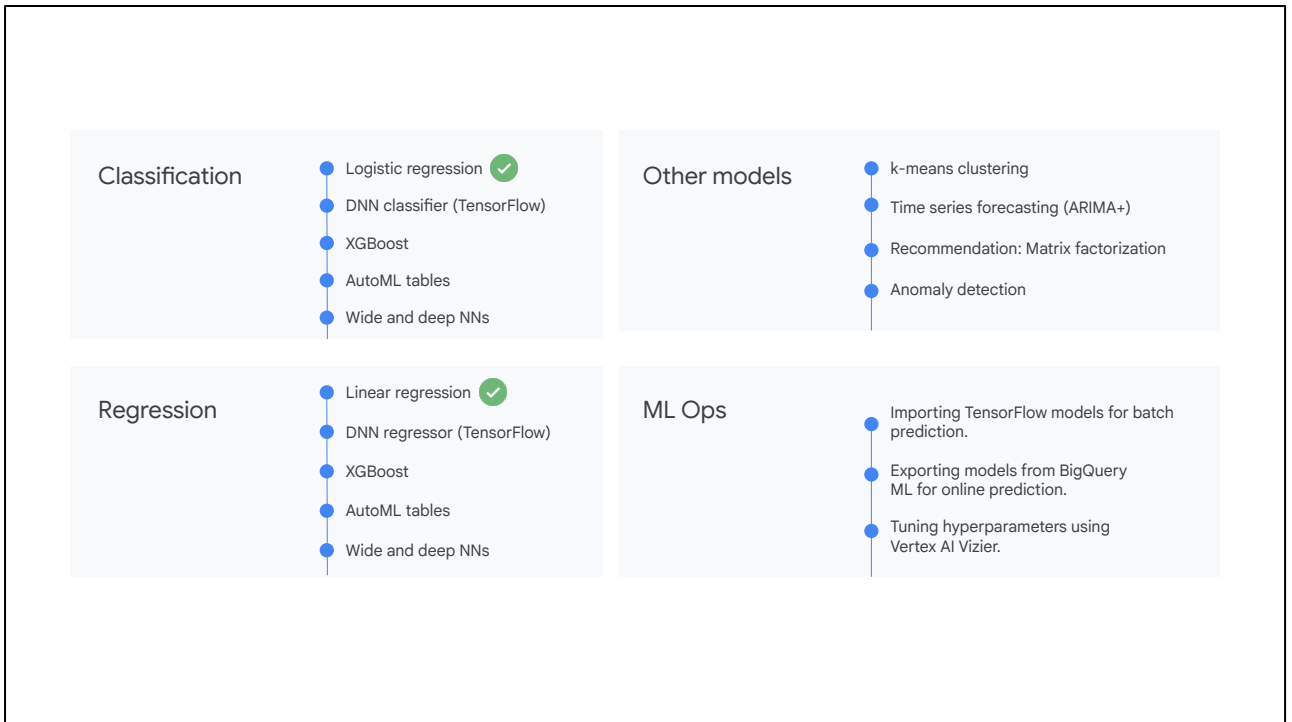
In *phase 3*, you *create the model* inside BigQuery.

This is done by using the `CREATE MODEL` command. In this example, you want to create an ML model to predict customer purchasing behavior, specifically if they will buy this product in the future.

You give the model a name: `ecommerce.classification`.

You then specify the model type. Remember the previous lesson about ML model types. If you want to predict whether a customer will buy or not, which ML model should you use?

That's right. A logistic regression model is the answer, because you are solving a classification problem.



Apart from the logistic regression model to solve the classification problem, BigQuery ML also supports other popular ML models.

They include regression models such as linear regression and other models, such as k-means clustering and time series forecasting models.

In addition to providing different types of machine learning models, BigQuery ML supports MLOps, machine learning operations. MLOps turns your ML experiment to production and helps deploy, monitor, and manage the ML models. You'll learn more about MLOps later in this course.

You are recommended to start with simple options such as logistic regression and linear regression, and use the results as a benchmark to compare against more complex models such as DNN (deep neural networks), which take more time and computing resources to train and deploy.

Key phases

01 Extract, transform, and load data into BigQuery.

02 Select and preprocess features.

03 Create the model inside BigQuery.

04 Evaluate the performance of the trained model.

05 Use the model to make predictions.

Use the `CREATE MODEL` command.

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) AS

# SQL query with training data
```

After specifying the model type, you also need to define the label column. Why?

Remember the two major categories of ML models, supervised and unsupervised? The former deals with labeled data and predicts a goal, whereas the latter handles unlabeled data and identifies a hidden pattern. Is this a supervised or unsupervised model?

Of course, it's a supervised classification problem, thus a labeled column.

From there, you can run the query.

Key phases

01 Extract, transform, and load data into BigQuery.

02 Select and preprocess features.

03 Create the model inside BigQuery.

04 Evaluate the performance of the trained model.

05 Use the model to make predictions.

Execute an `ML.EVALUATE` query.

```
#standardSQL
SELECT
  roc_auc,
  accuracy,
  precision,
  recall
FROM
  ML.EVALUATE(MODEL
    `ecommerce.classification`)
# SQL query with evaluation data
```

In *phase 4*, after your model is trained, you can execute an `ML.EVALUATE` query to *evaluate the performance of the trained model* on your evaluation dataset.

It's here that you specify which evaluation metrics the model will access, such as accuracy, precision, and recall. You'll explore these metrics later in the next module.

Key phases

01 Extract, transform, and load data into BigQuery.

02 Select and preprocess features.

03 Create the model inside BigQuery.

04 Evaluate the performance of the trained model.

05 Use the model to make predictions.

Invoke the ``ML.PREDICT`` command.

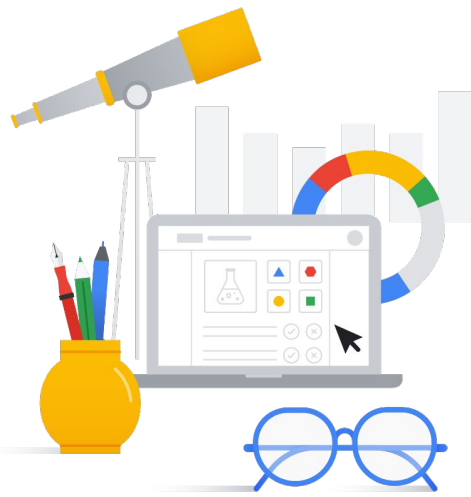
```
#standardSQL
SELECT * FROM
  ML.PREDICT
  (MODEL
   `ecommerce.classification`)
```

SQL query with test data

Finally, in *phase 5*, when you're happy with your model performance, you can then use it to *make predictions*.

To do so, invoke the ``ML.PREDICT`` command on your newly trained model to return with predictions and the model's confidence in those predictions.

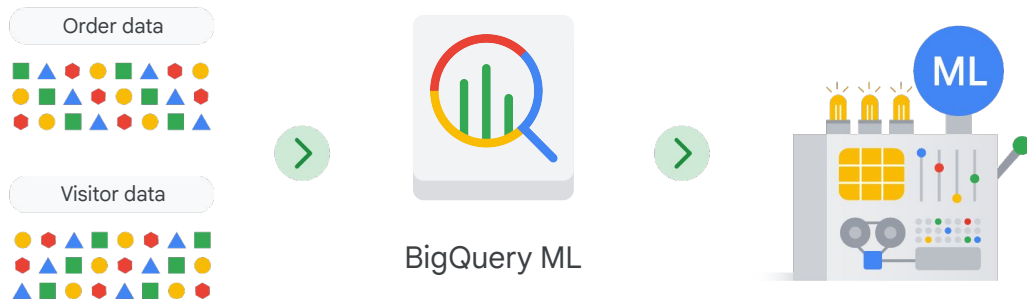
With the results, your label field will have "predicted" added to the field name. This is your model's prediction for that label.



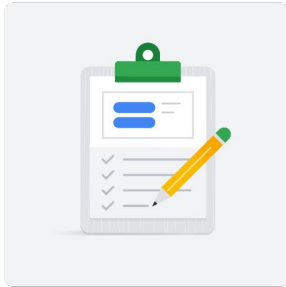
Hands-on lab:

Predict Visitor Purchases
with BigQuery ML

Ready for hands-on practice? Let's apply these steps and build your first ML model in BigQuery!



In the upcoming lab, you'll use real ecommerce data from the Google Merchandise Store to predict whether a visitor will make future purchases.



BigQuery ML lab

- 1 Create a dataset in BigQuery.
- 2 Train an ML model.
- 3 Evaluate the model.
- 4 Use the model for prediction.

You'll gain valuable experience creating datasets, training and evaluating ML models, and using them for predictions.



SQL



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01 AI foundations



02 Generative AI



03 AI development options



04 AI development workflow



Congratulations on completing the first module, AI foundations!

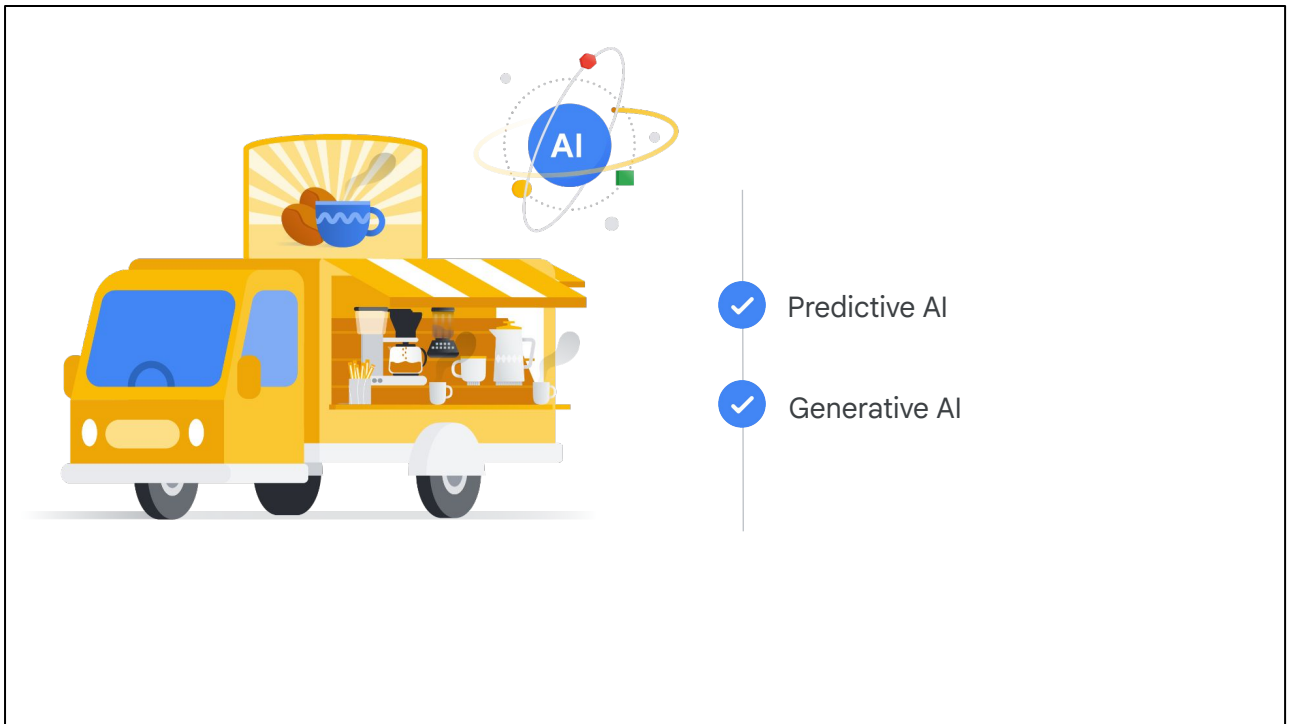
What are three key terms you learned in this module?

1

2

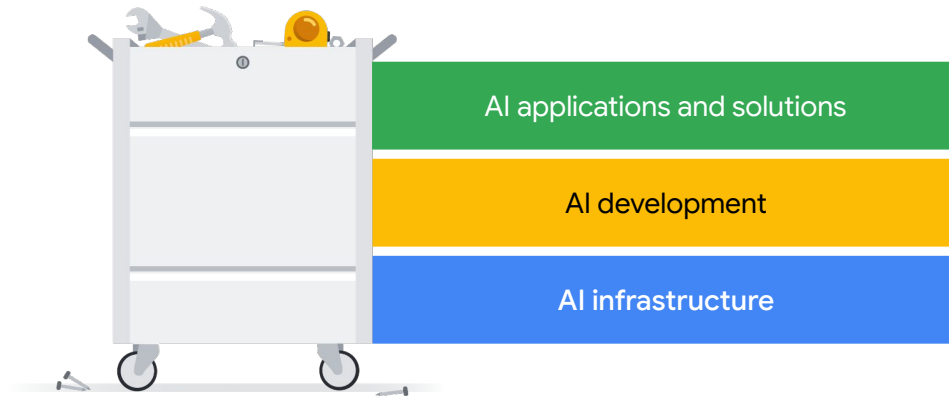
3

Take a moment to reflect and write down what you believe are three key terms you learned from the module.



Instead of starting with a morning coffee, you kicked off your journey with Coffee on Wheels, a fascinating food truck use case.

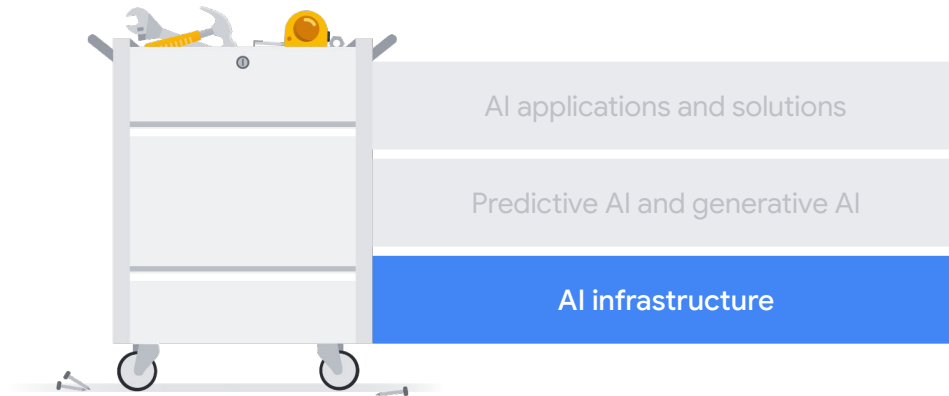
It brilliantly showcased how AI is evolving from a mere efficiency tool to a powerhouse of innovation, tackling both traditional predictive AI and modern generative AI challenges, tackling both traditional predictive AI, and modern generative AI challenges.



To help you achieve your AI vision, you were then introduced to the AI/ML toolbox on Google Cloud, which is structured into three layers:

- AI infrastructure,
- AI development,
- and AI applications and solutions.

This course focuses on the second layer, exploring both predictive AI and generative AI in the following modules.



Before that, you dove into the foundational layer of AI infrastructure, specifically compute and storage. You even discovered the data and AI products that guide the journey from data to AI, covering ingestion and processing, storage and analysis, and finally, activation with AI.

Supervised learning



Labeled data, task-driven,
identifies a goal.

Unsupervised learning



Unlabeled data, data-driven,
identifies a pattern.

Before diving into practical applications, you were equipped with AI model basics, focusing on supervised versus unsupervised learning.

Key phases

01 Extract, transform, and load data into BigQuery.

02 Select and preprocess features.

03 Create the model inside BigQuery.

04 Evaluate the performance of the trained model.

05 Use the model to make predictions.



BigQuery

With these concepts in mind, you rolled up your sleeves and honed in on the module's highlight: BigQuery ML.

Here, you explored various models and walked through the steps to build your own ML model.



SQL



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01 AI foundations



02 Generative AI



03 AI development options



04 AI development workflow



With these AI foundations firmly in place, we'll move on to the most recent advancements in generative AI in the next module. See you soon!